

Bayesian Age-Period-Cohort Modeling

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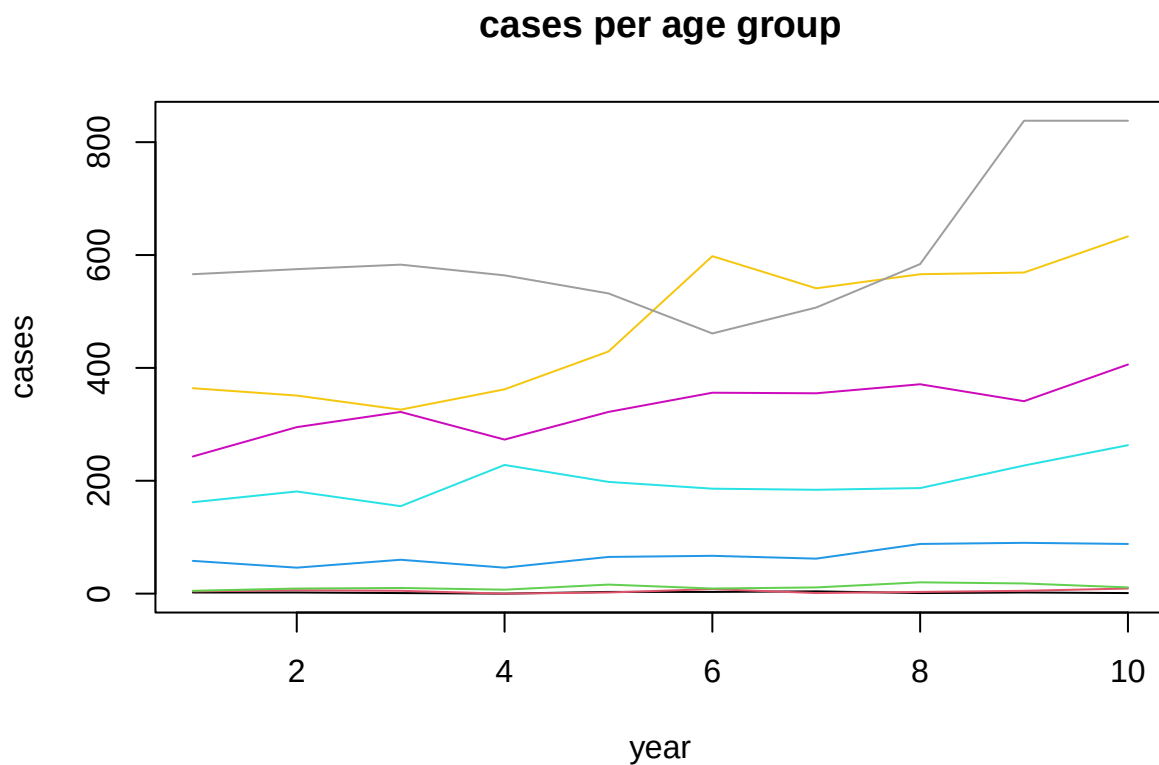
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This vignette walks through the range of age-period-cohort (APC) models that `bamp()` can fit: different smoothness priors for the age, period and cohort effects, a model that omits an effect entirely, and models where the period or cohort effect is scaled by a known covariate. For a first introduction to `bamp()`, see *vignette("bamp", package="bamp")*; for forecasting, see *vignette("prediction", package="bamp")*.

Data example

BAMP includes a data example: `cases` and `population`, matrices of observed case counts and population at risk, cross-classified by calendar year (rows) and age group (columns). It also bundles `cov_p` and `cov_c`, example period and cohort covariates used in the last two models below.

```
data(apc)
plot(cases[,1],type="l",ylim=range(cases), ylab="cases", xlab="year", main="cases per age group")
for (i in 2:8)lines(cases[,i], col=i)
```



APC model with random-walk first-order priors

A first-order random walk ("rw1") prior assumes each effect changes smoothly, without a systematic trend, between neighbouring age groups, periods or cohorts. This is the most common default.

```
model11 <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1",  
               periods_per_agegroup = 5)
```

bamp() automatically checks MCMC convergence using the Gelman-Rubin diagnostic. We can also run this check manually:

```
checkConvergence(model11)
```

```
## [1] TRUE
```

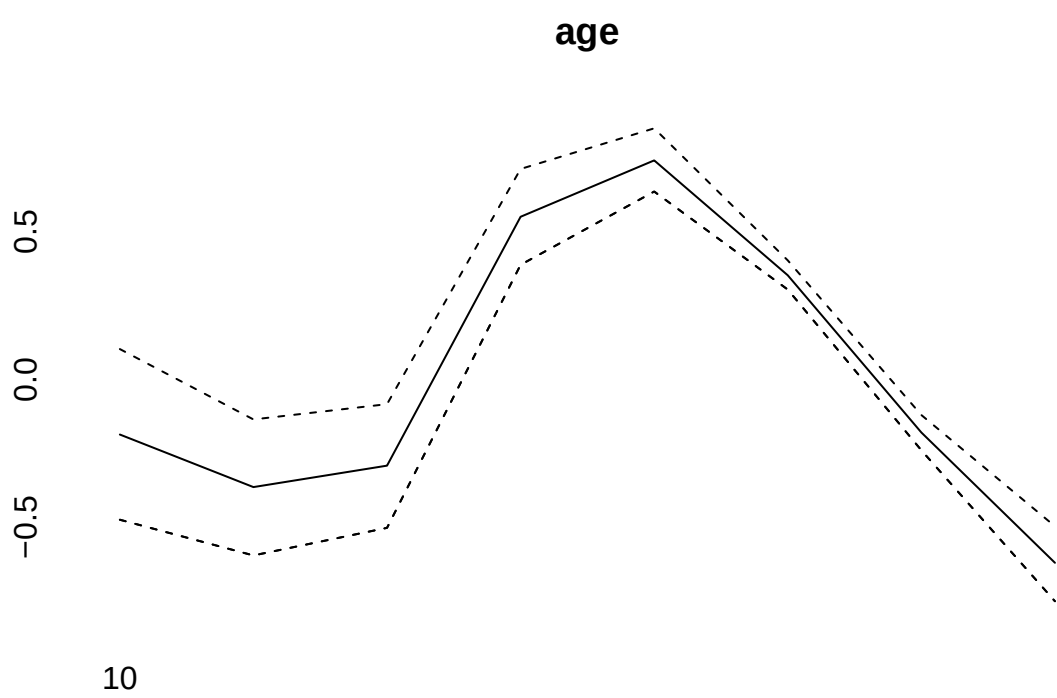
print() summarises the fitted model: posterior estimates of the smoothing (precision) parameters, and the deviance and DIC, which are useful for comparing the models fitted in this vignette:

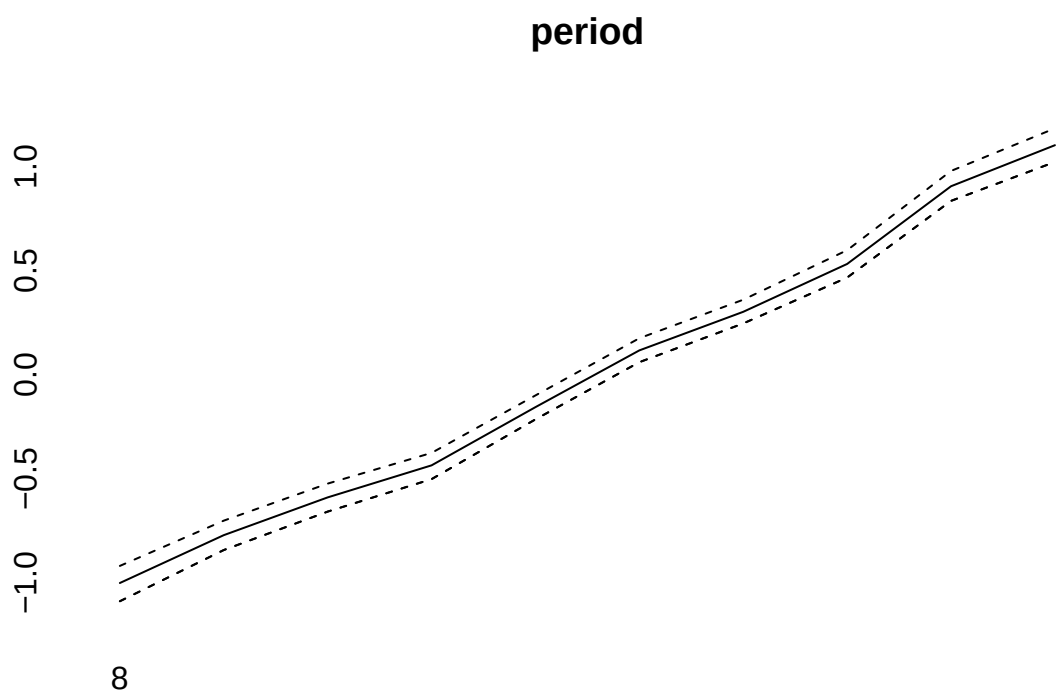
```
print(model11)
```

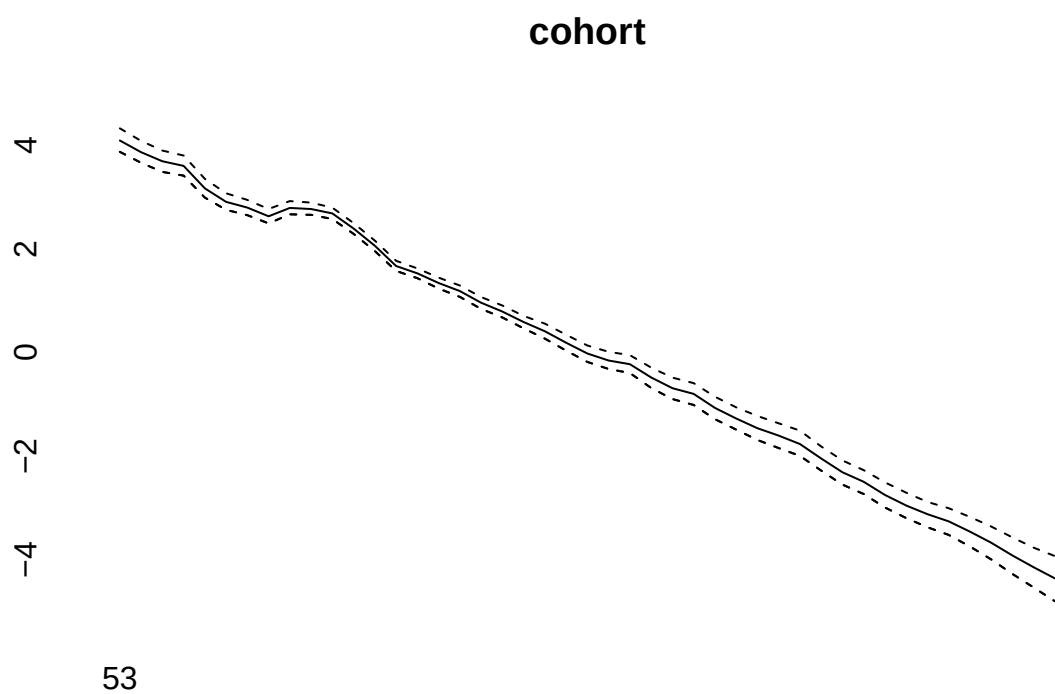
```
##  
## Model:  
## age (rw1) - period (rw1) - cohort (rw1) model  
## Deviance:      229.97  
## pD:            36.53  
## DIC:           266.50  
##  
##  
## Hyper parameters:           5%           50%           95%  
## age                        0.439         1.104         2.318  
## period                     49.454        128.729        322.369  
## cohort                     34.853         60.676        102.902  
##  
##  
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

The fitted age, period and cohort effects can be plotted with point-wise posterior quantiles:

```
plot(model11)
```



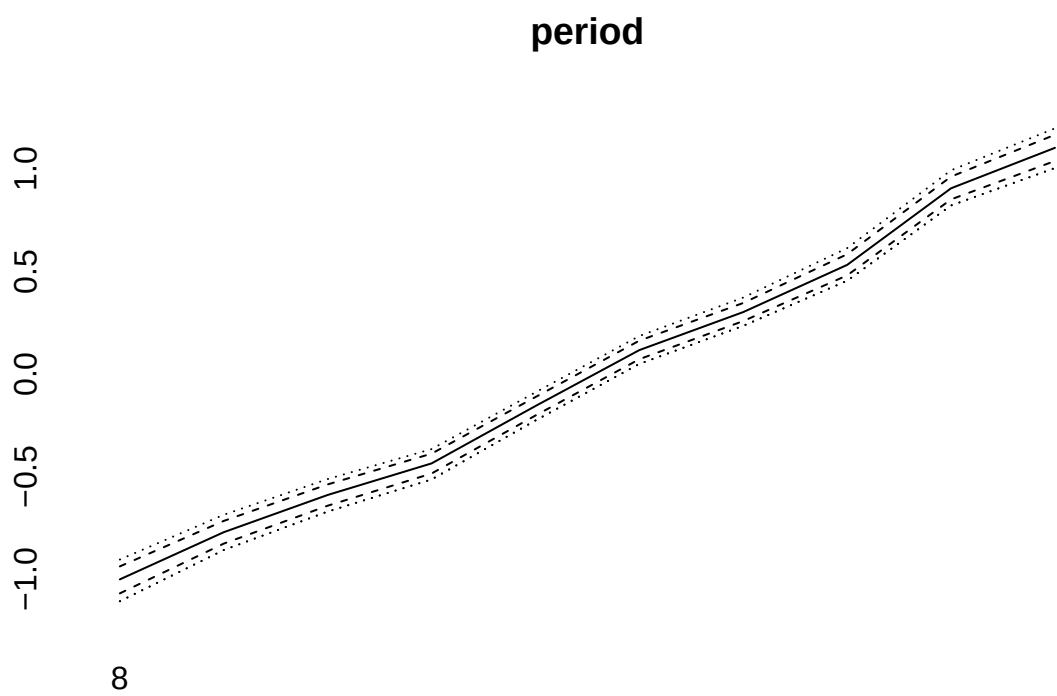


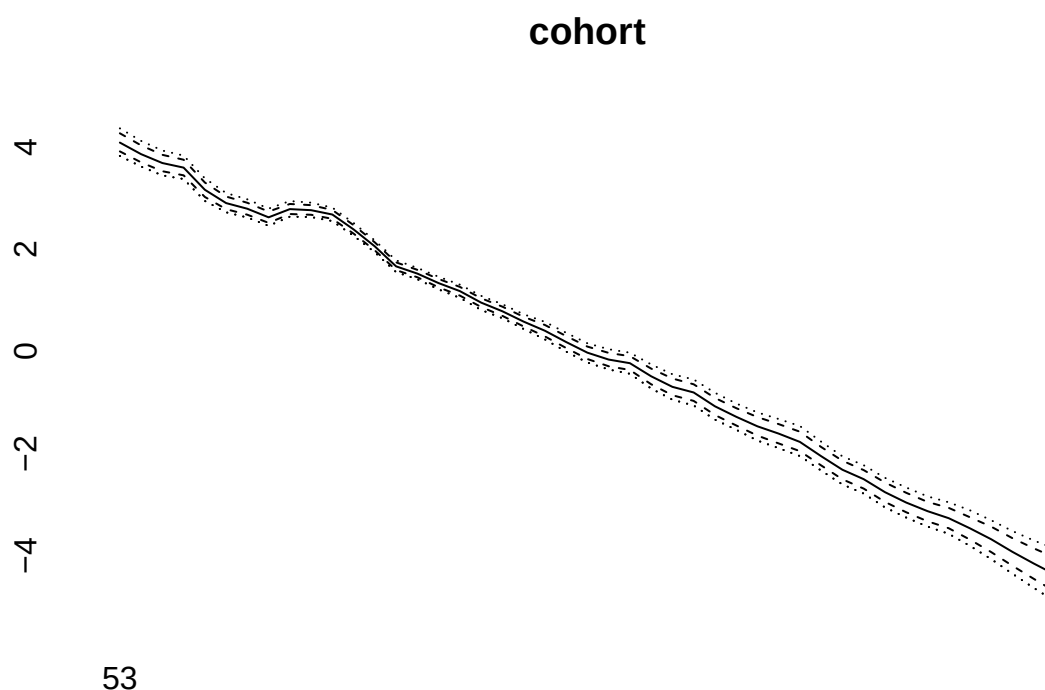


By default the plot shows the median and a 90% interval; other quantiles can be requested explicitly:

```
plot(model1, quantiles = c(0.025,0.1,0.5,0.9,0.975))
```







APC model with random-walk second-order priors

A second-order random walk ("rw2") prior penalises curvature rather than level, so it produces effects that follow a smooth, roughly linear trend instead of reverting towards a constant. RW2 priors are less informative than RW1, so the chains mix more slowly and need substantially more iterations to converge; here we also set the smoothing hyperparameters explicitly instead of using the defaults.

```
model2 <- bamp(cases, population, age="rw2", period="rw2", cohort="rw2",
               periods_per_agegroup = 5,
               mcmc.options=list("number_of_iterations"=200000, "burn_in"=100000, "step"=50, "tuning"=50),
               hyperpar=list("age"=c(1,.5), "period"=c(1,0.05), "cohort"=c(1,0.05)))
```

```
checkConvergence(model2)
```

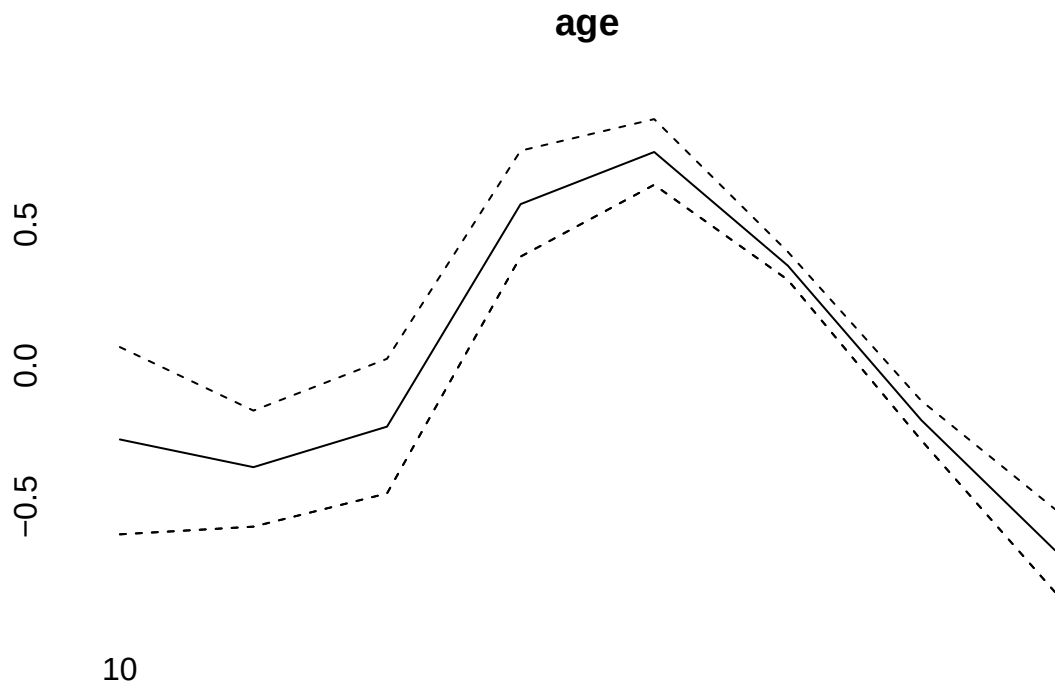
```
## [1] TRUE
```

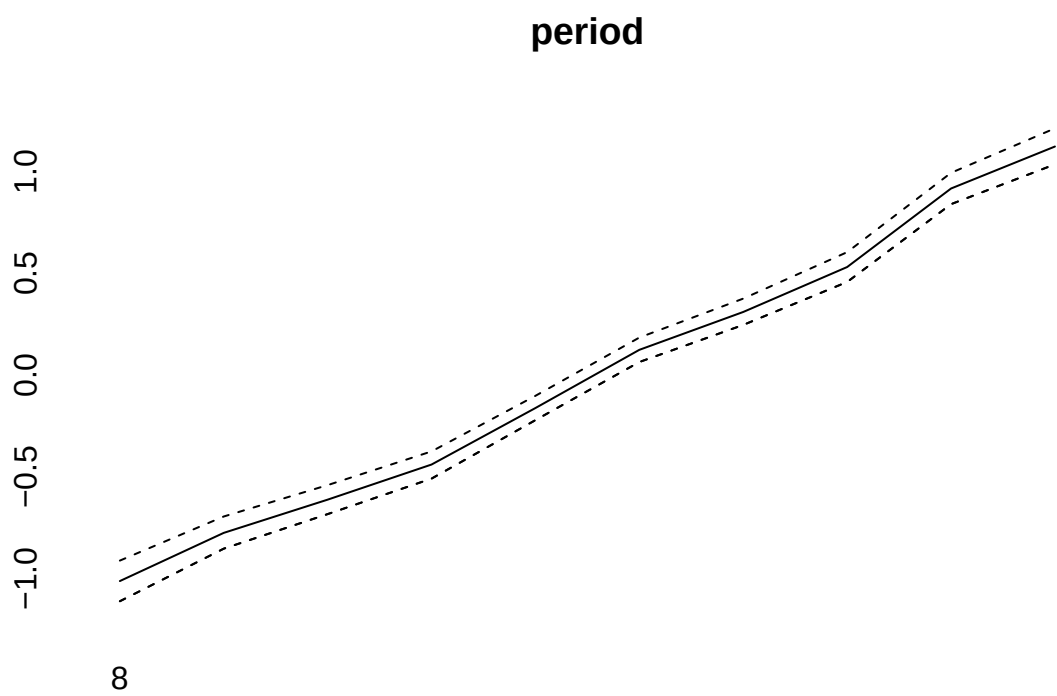
```
print(model2)
```

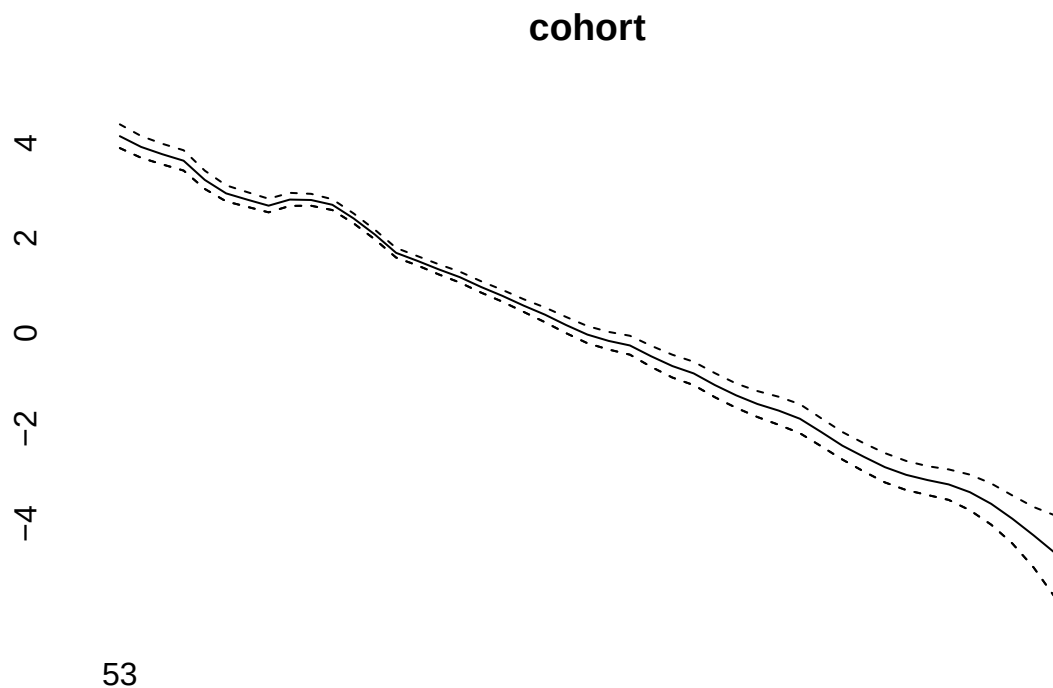
```
##
## Model:
## age (rw2) - period (rw2) - cohort (rw2) model
## Deviance:    233.68
## pD:          35.79
## DIC:         269.47
##
##
## Hyper parameters:
## age          5%          50%          95%
##              1.099      3.079      6.928
```



```
## period          15.683      41.174      89.223
## cohort          24.730      46.665      84.145
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
plot(model2)
```







APC model without a period effect

Age, period and cohort are linearly dependent – any one is determined by the other two – so a model with all three as unrestricted smooth trends is not identifiable. One common way to sidestep this is to drop the effect of least interest – here the period effect (`period=" "`) – and keep only age and cohort:

```
model3<-bamp(cases, population, age="rw1", period=" ", cohort="rw2",
             periods_per_agegroup = 5)
```

```
checkConvergence(model3)
```

```
## [1] TRUE
```

```
print(model3)
```

```
##
```

```
## Model:
```

```
## age (rw1) cohort (rw2) model
```

```
## Deviance:      276.08
```

```
## pD:            29.90
```

```
## DIC:           305.98
```

```
##
```

```
##
```

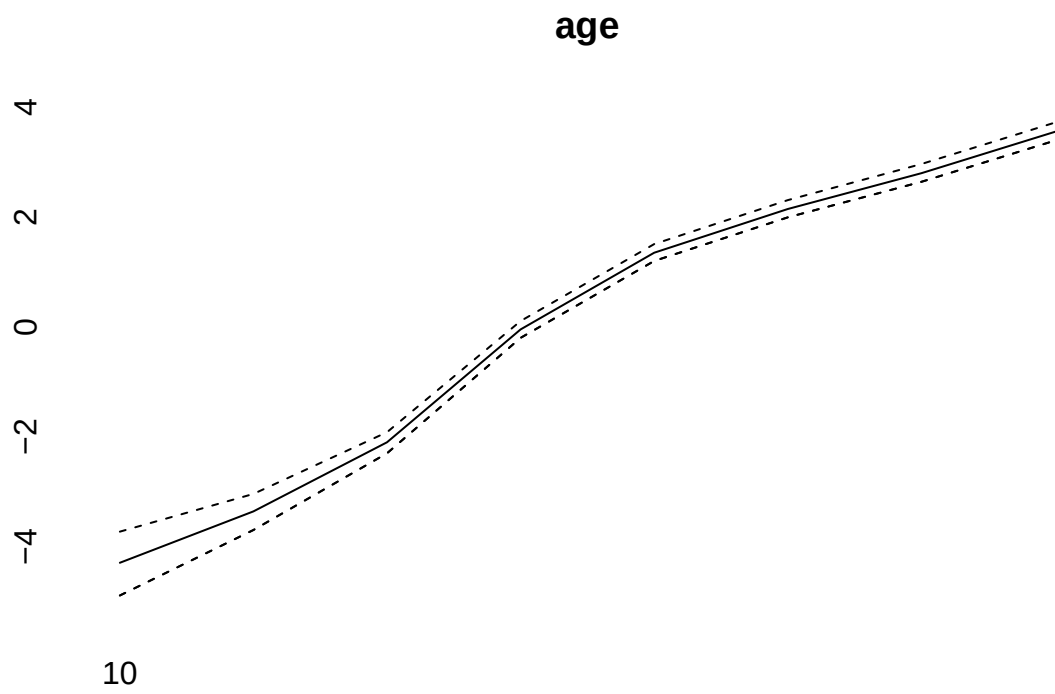
```
## Hyper parameters:
```

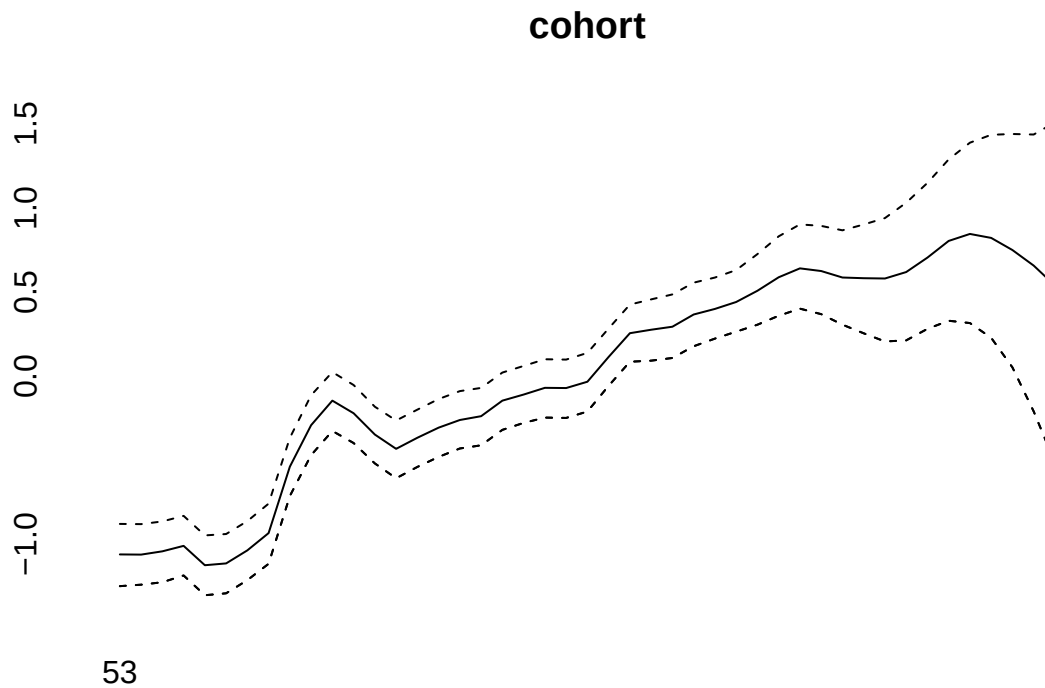
	5%	50%	95%
## age	0.290	0.733	1.556
## cohort	37.980	72.712	137.381

```
##
```

```
##
```

```
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).  
plot(model3)
```





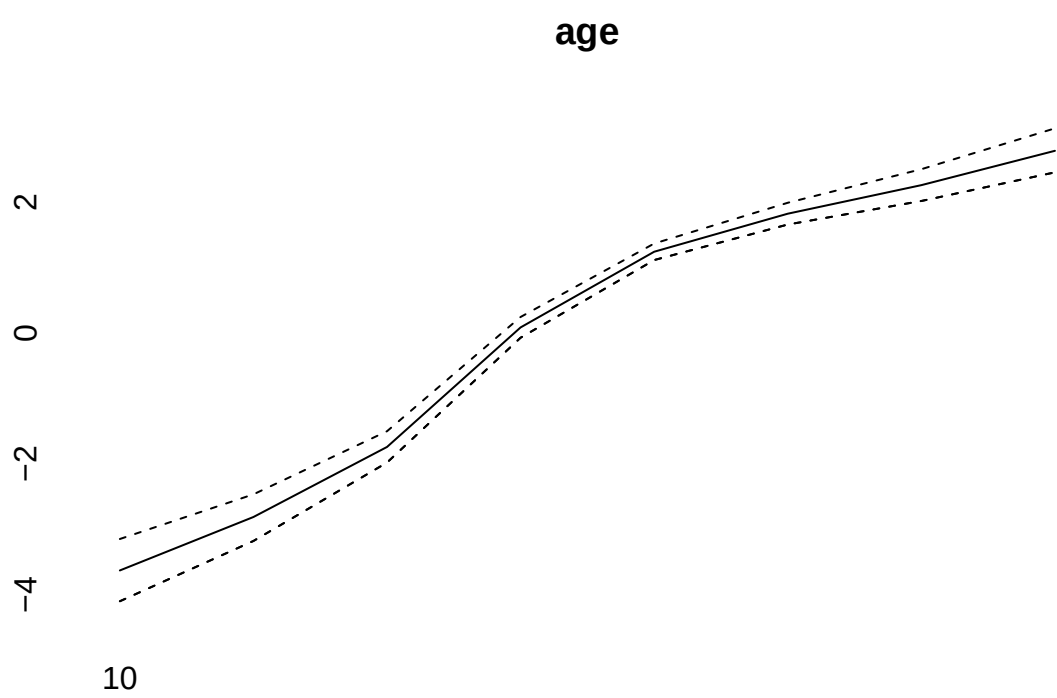
APC model with a cohort covariate

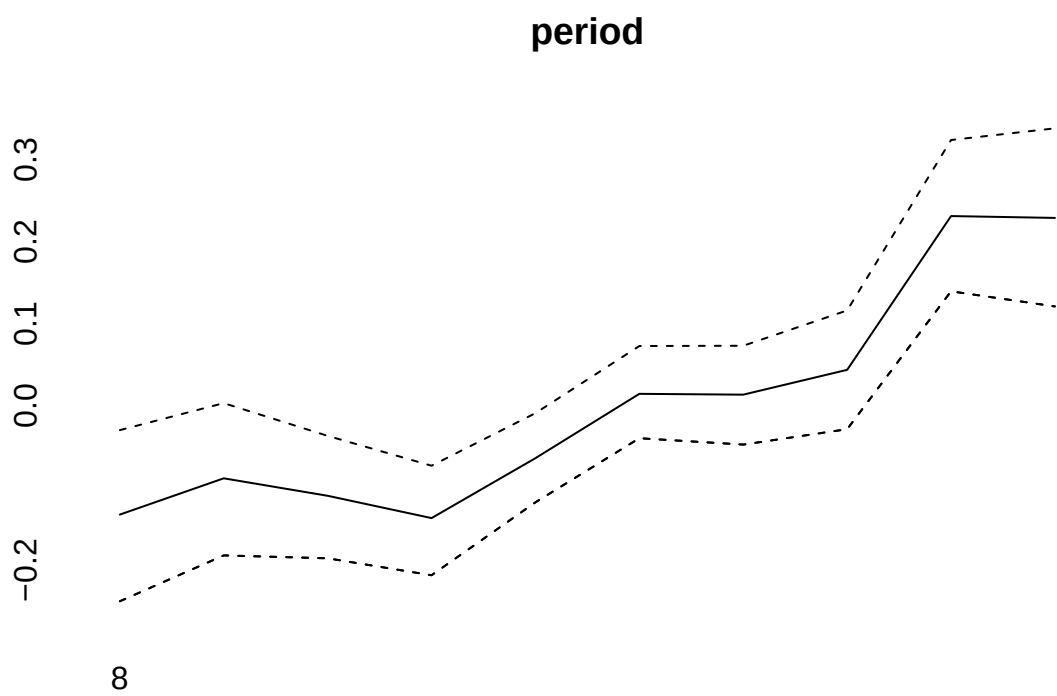
Instead of dropping an effect, a known covariate can scale it: `cohort_covariate` multiplies the cohort effect by a given value at each cohort (here the bundled example covariate `cov_c`), which is useful when an external, cohort-specific risk factor is available.

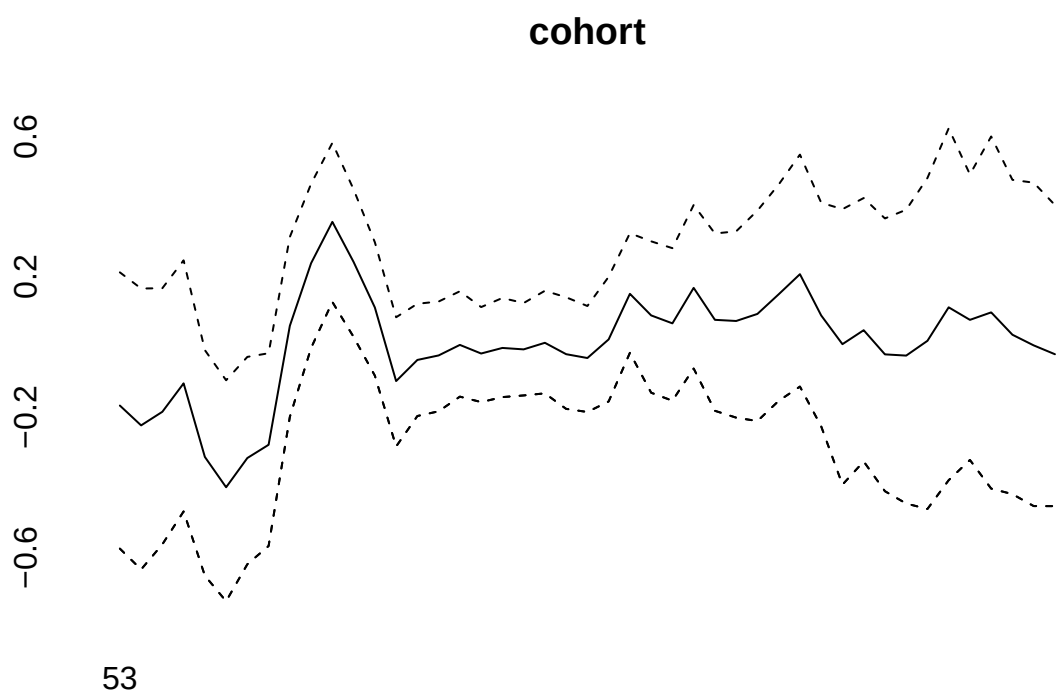
```
(model4<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  cohort_covariate = cov_c, periods_per_agegroup = 5))
```

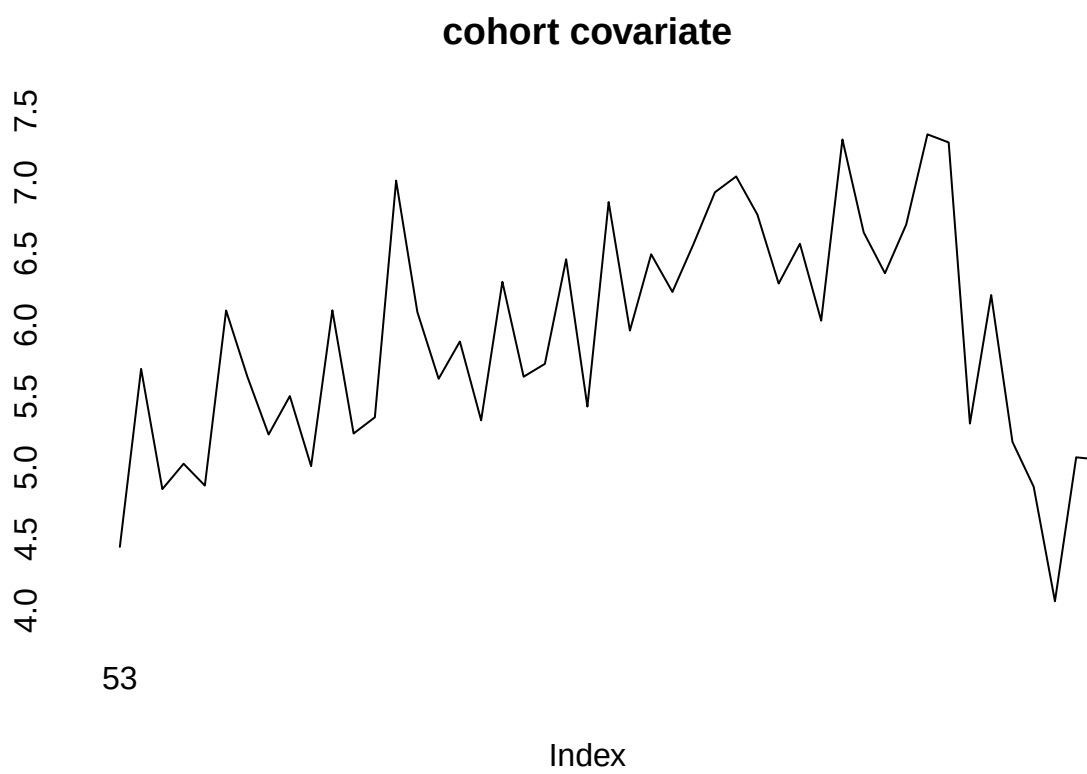
```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      230.37
## pD:            37.51
## DIC:           267.88
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.407         1.015         2.099
## period                     55.079        143.819        339.471
## cohort                     31.066         53.568         91.340
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model4)
```

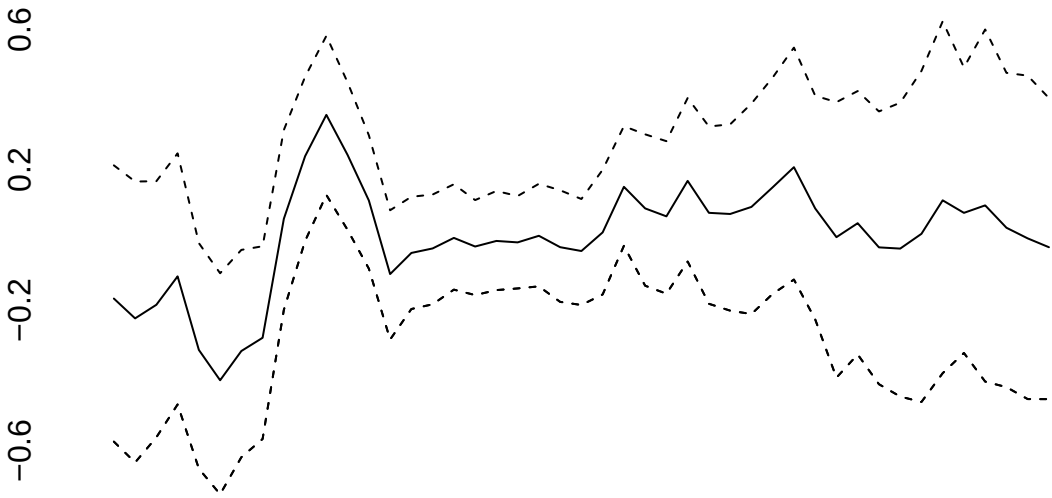




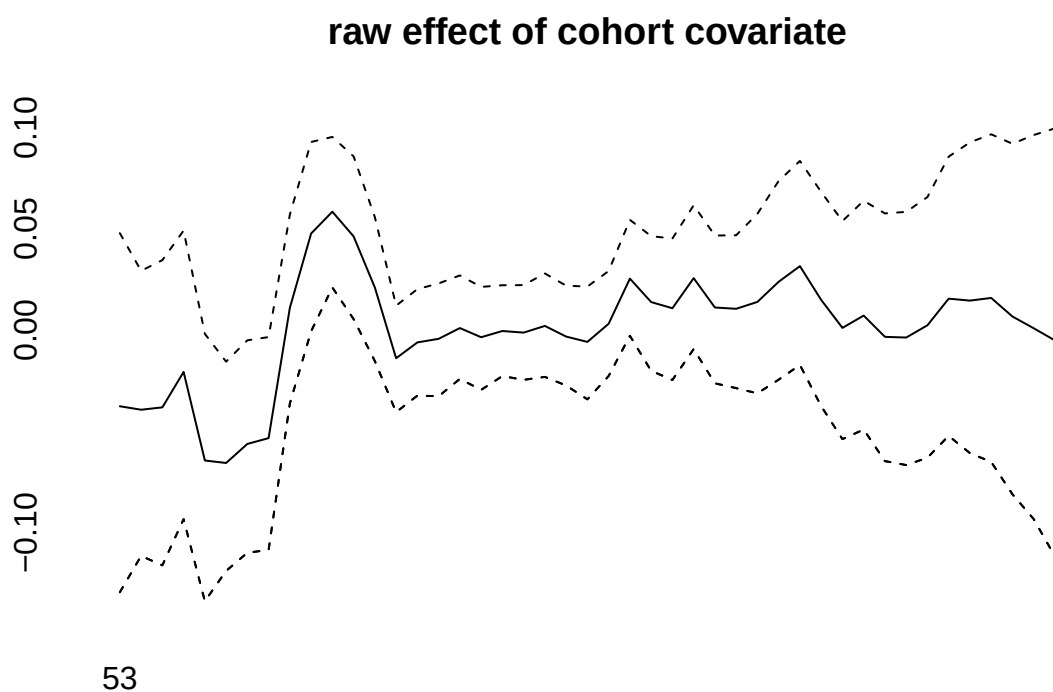




cohort effect



53



APC model with a period covariate

Analogously, `period_covariate` scales the period effect by a known, period-specific covariate (here `cov_p`):

```
(model15<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  period_covariate = cov_p, periods_per_agegroup = 5))
```

```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      229.28
## pD:            36.60
## DIC:           265.88
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.428         1.082         2.266
## period                     54.466        148.829        370.075
## cohort                     34.836         59.178         96.785
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model5)
```

